Covariant Formulation of Shock Waves Propagation for Traffic Flow on Curved Roads

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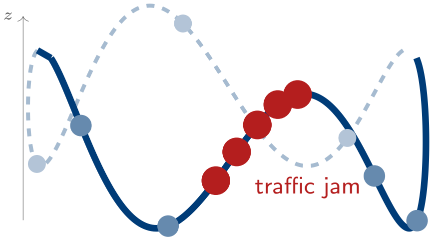
Traffic jams on curved roads

Traffic waves propagate on real road geometry
— curves, hills, ramps —
but models assume **flat space**.

- Empirical evidence: slower flow at sharp turns, jam accumulation on inclines
- Highway Capacity Manual adjusts capacity for curvature — but *ad hoc*
- No first-principles coupling of geometry to shock dynamics



3D road with spatially varying geometry.
Vehicles cluster where curvature is high.

Phantom jams: no bottleneck needed

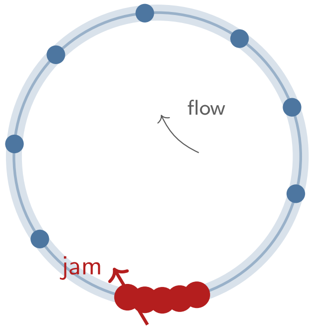
Sugiyama *et al.* (2008) [1]:

22 cars on a circular track. Uniform initial speed. **No traffic lights. No accidents.**

Within minutes: a spontaneous traffic jam propagating *backward* against the flow.

Even on a **flat circle** the system is unstable to density waves.

Question: What happens on a road with spatially varying geometry?



Prior work and the gap we address

What exists:

- Conservation laws on Riemannian manifolds: well-posedness, FV schemes [2,3]
- Traffic on curved roads: lattice and macro models with curvature/slope effects [4,5,6]
- KdV–Burgers on curved manifolds: Laplace–Beltrami dissipation, shock profiles [7]
- Embedding methods for conservation laws on manifolds in $\mathbb{R}^3$ [8]

What is missing:

The gap

No published model combines:

1. A macroscopic traffic model (LWR)
2. on a spatial curve $\gamma(s) \in \mathbb{R}^3$
3. with explicit $\kappa(s), \tau(s)$ in the fundamental diagram
4. and derives how geometry modifies the **shock velocity**

This is what we do.

Spatial curves in $\mathbb{R}^3$: the Frenet–Serret frame

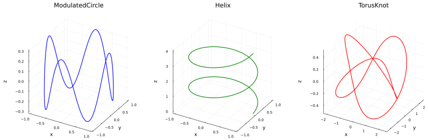
A road is a smooth closed curve $\gamma(s) \in \mathbb{R}^3$ parametrised by arc length s.

The **Frenet–Serret apparatus**:

$$\mathbf{T}'(s) = \kappa(s)\,\mathbf{N}(s)$$
$$\mathbf{N}'(s) = -\kappa(s)\,\mathbf{T}(s) + \tau(s)\,\mathbf{B}(s)$$

- $\kappa(s)$: curvature (how sharply the road turns)
- $\tau(s)$: torsion (how the road twists out of plane)

These are the **intrinsic geometric data** that should couple to flow dynamics.



Four test geometries: straight line ($\kappa = 0, \tau = 0$), circle ($\kappa =$const, $\tau = 0$), helix ($\kappa, \tau =$const), modulated 3D curve (κ varies $\sim 800\times$).

Covariant scalar conservation law

The Lighthill–Whitham–Richards (LWR) equation on a **curved road**:

Covariant LWR equation

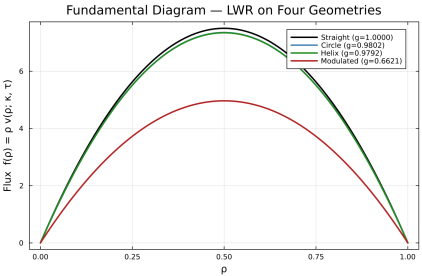
In angular coordinates φ: $\quad \partial_t(\rho\, h) + \partial_\varphi(\rho\, v) = 0$

In arc-length s: $\quad \partial_t \rho + \partial_s[\rho\, v(\rho;\kappa,\tau)] = 0$

The flux depends on geometry via a **fundamental diagram**. We *model*:

$$v(\rho;\kappa,\tau) = v_{\max}\, g(\kappa,\tau)\left(1 - \frac{\rho}{\rho_{\max}}\right)$$

with the geometric correction (phenomenological closure):



Fundamental diagrams (LWR) for the four geometries. Curvature reduces max flux by 2.0%; torsion adds 0.07%.

Rankine–Hugoniot conditions on curved geometry

At a shock discontinuity, the jump condition gives the **shock velocity**:

Geometric Rankine–Hugoniot

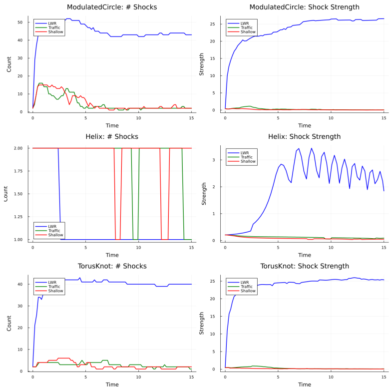
$$V_{\text{shock}}(s) = \frac{f(\rho_R; s) - f(\rho_L; s)}{\rho_R - \rho_L}$$

where $f(\rho; s) = \rho\, v(\rho; \kappa(s), \tau(s))$

For **constant geometry** (κ, τ uniform):

- $V_{\text{shock}} = v_{\max}\, g\,(1 - \rho_L - \rho_R)$
- Exact prediction: no free parameters

For **variable geometry**: $V_{\text{shock}}(s)$ depends on position — the shock accelerates and decelerates as it traverses the road.



Shock analysis across geometries. For constant κ, τ: theory matches numerics to machine precision.

The effective potential: why $\partial_s \kappa$ matters

Expanding $\partial_s f$ in the conservation law:

$$\partial_t \rho + \lambda(\rho, s)\, \partial_s \rho = \underbrace{-\rho\, v(\rho)\, \partial_s \Phi}_{\text{geometric source}}$$

Effective potential

$$\Phi(s) = \beta_\kappa |\kappa(s)| + \beta_\tau |\tau(s)|$$

Physical interpretation:

- $\partial_s \Phi > 0$ (increasing curvature): density accumulates
- $\partial_s \Phi < 0$ (decreasing curvature): density depletes
- $\partial_s \Phi = 0$ (constant geometry): **no geometric forcing**

This explains **three key observations**:

1. **Helix** ($\partial_s \Phi = 0$): no jammitons
2. **Modulated curve** ($\partial_s \Phi$ large): density traps in high-κ regions
3. Shocks refract across Φ-gradients — analogous to Snell's law

Godunov finite-volume scheme

We solve the covariant LWR equation with a **Godunov scheme** on the arc-length coordinate s:

1. Piecewise-constant reconstruction on each cell
2. Exact Riemann solver at each interface: $f^* = f(\rho^*)$ with ρ^* from the local Rankine–Hugoniot condition
3. Conservative update:

$$\rho_j^{n+1} = \rho_j^n - \frac{\Delta t}{\Delta s}\left(f^*_{j+1/2} - f^*_{j-1/2}\right)$$

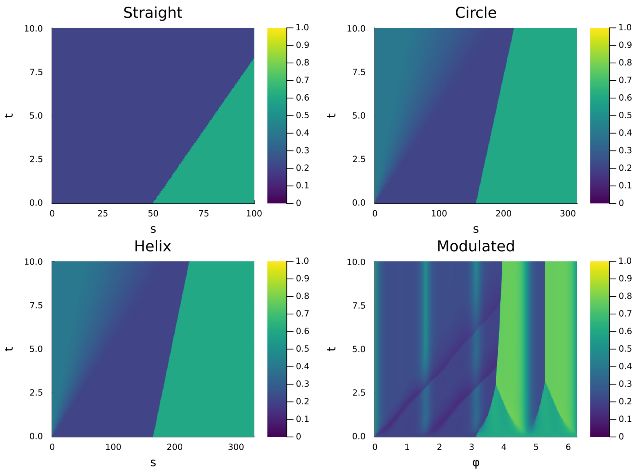
Mass conservation

Implementation:

- Julia package, ∼2000 lines
- $N_s = 500$ spatial cells
- CFL: $\Delta t \leq \Delta s / \max |f'|$
- Periodic BCs
- Riemann solver uses local $\kappa(s_j), \tau(s_j)$

Also implemented: MUSCL reconstruction with TVD limiters (minmod, van Leer, MC) for second-order accuracy.

Kymographs: shock propagation across geometries



Space-time density maps $\rho(s,t)$ for the four test geometries. Riemann problem: $\rho_L = 0.2 \to \rho_R = 0.6$.

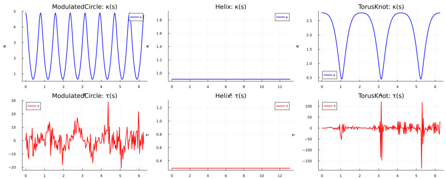
Constant geometry: uniform shock velocity. Variable geometry: shock speed modulated by local $\kappa(s)$.

Constant-geometry validation

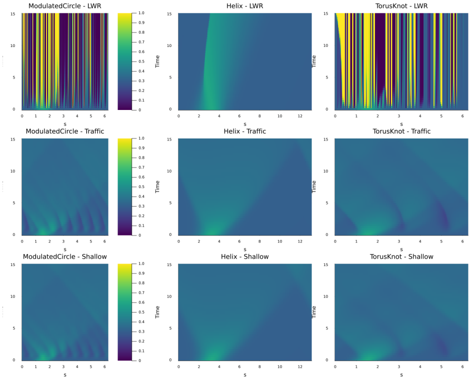
For the three constant-geometry cases ($v_{\max} = 30$, $\beta_\kappa = 1$, $\beta_\tau = 0.5$):

Geometry	κ	τ	V_{shock}
Straight	0	0	6.000 (ref.)
Circle ($R\!=\!50$)	0.02	0	5.881
Helix	0.02	0.002	5.877

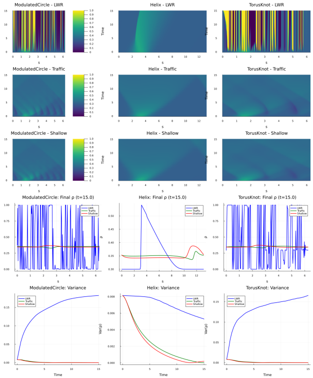
- For constant κ, τ the Riemann problem has an exact solution; the scheme recovers it to $< 10^{-15}$
- Curvature alone: $\mathbf{2.0}\%$ slowdown
- Adding torsion: additional $\mathbf{0.07}\%$
- These validate the scheme; the interesting



$\kappa(s)$ and $\tau(s)$ profiles. The modulated curve has κ varying by $\sim 800\times$.



Kymographs: shock front bends across regions of different κ.



Density evolution, shock trajectories, and velocity modulation on the modulated curve.

Modulated circle: $\kappa \in [0.05, 40]$, velocity field $v \in [0, 13.6]$ — **1700% variation**. Shocks **nearly halt** at high κ and **accelerate** through low-κ stretches. This is the geometric refraction predicted by the effective potential $\Phi(s)$.

Connection to microscopic dynamics

The covariant LWR sits atop a **three-scale hierarchy** that we have verified independently:

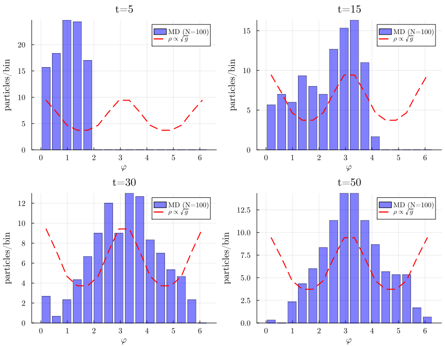
Microscopic
N hard rods on $\gamma(s)$
event-driven MD

$\downarrow N \to \infty$

Mesoscopic
Boltzmann/GHD on curve
covariant Burgers eq.

$\downarrow$ moments

Macroscopic
$\partial_t \rho + \partial_s(\rho v) \;=\; 0$
covariant LWR



Lines: covariant Burgers PDE (mesoscopic level). **Symbols**: event-driven MD, 500 hard rods on an ellipse. 24 initial conditions, no free parameters.

The Christoffel term Γu^2 is validated microscopically. Traffic LWR is the

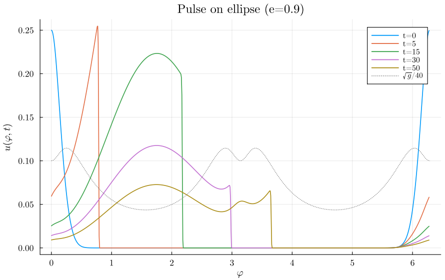
Arc-length universality

Key theoretical property:

In arc-length s the metric is 1 by construction, so the equation reduces to **flat-space LWR** with position-dependent fundamental diagram. The Christoffel term Γu^2 is absorbed into the coordinate transformation.

All exact solutions of flat Burgers *transplant* to the curved road:

- N-wave $\to$ metric-distorted N-wave
- Sawtooth $\to$ metric-modulated sawtooth
- Cole–Hopf viscous profile $\to$ curved Cole–Hopf



N-wave on an ellipse: the two fronts travel at different speeds — the pulse distorts asymmetrically.

Consequence

Summary and outlook

What we showed:

- Covariant LWR on curves in $\mathbb{R}^3$ with $\kappa(s), \tau(s)$ in the flux
- Effective potential $\Phi(s)$ governs density redistribution
- Godunov: mass error $< 10^{-15}$
- Constant geometry: exact V_{shock}
- Variable geometry: geometric refraction

Future work:

- Agent-based IDM validation on curves
- GPS data on curved highways
- Active matter on manifolds
- Higher-order MUSCL/WENO schemes
- Networks of curved roads

Geometry	V_{shock}	**Error vs. R–H**	**Effect**
Straight ($\kappa = 0$)	6.000	0 (ref.)	—
Circle ($\kappa =$const)	5.881	$< 10^{-15}$	-2.0%
Helix ($\kappa, \tau =$const)	5.877	$< 10^{-15}$	-2.07%
Modulated (κ variable)	varies	pos.-dependent	geom. refraction

Thank you!

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Questions?

References

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Backup: Simulation parameters

The geometric correction to the fundamental diagram accounts for the reduced effective speed on curves:

A vehicle on a curve of curvature κ experiences centripetal acceleration $a_c = v^2\kappa$. Stability requires $a_c \le \mu g_{\mathrm{grav}}$, giving the speed reduction:

$$v_{\max}(\kappa) = v_{\max}^{\mathrm{flat}}\sqrt{\frac{\mu g_{\mathrm{grav}}}{\kappa}} \quad \text{for large } \kappa$$

The exponential model $g = e^{-\beta_\kappa|\kappa|}$ interpolates smoothly between flat ($\kappa = 0$) and highly curved regimes.

Parameters used:

- $v_{\max} = 30.0$
- $\rho_{\max} = 1.0$
- $\beta_\kappa = 1.0$ (curvature sensitivity)
- $\beta_\tau = 0.5$ (torsion sensitivity)
- $\alpha = 1$ (Greenshields)
- Riemann: $\rho_L = 0.2,\ \rho_R = 0.6$

Test geometries:

- Circle: $R = 50,\ \kappa = 1/R,$ $\tau = 0$
- Helix: $R = 50,\ h = 100$
- Modulated: $\gamma(\varphi) =$ $(a\cos\varphi, a\sin\varphi, a^2\cos 2\varphi),$

Backup: Covariant Burgers equation

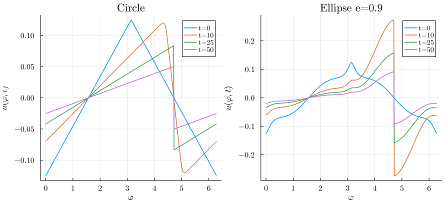
From the kinetic hierarchy (moment closure), the velocity field satisfies the **covariant Burgers equation**:

$$\partial_t u + u\,\partial_\varphi u + \Gamma(\varphi)\,u^2 = 0$$

where $\Gamma = g'/(2g)$ is the Christoffel symbol of the metric $g(\varphi) = |\gamma'(\varphi)|^2$.

Key property: in arc-length coordinates s, the physical velocity $v = \sqrt{g}\,u$ satisfies the flat Burgers equation.

This establishes exact equivalence between curved and flat dynamics up to coordinate transformation.



N-wave distortion on the ellipse. The Christoffel term Γu^2 accelerates the wave in low-g regions and decelerates it in high-g regions.